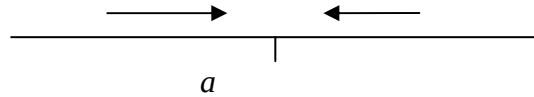


§2-2—2-3 Limit

Homework : §2-2 6,9,12,19,23,25,32,38

§2-3 2,11,17,21,29,39,46,49,58,59

$$\lim_{x \rightarrow a} f(x) = L, \quad \lim_{x \rightarrow a^+} f(x) = L \text{ (右極限)}, \quad \lim_{x \rightarrow a^-} f(x) = L \text{ (左極限)}$$

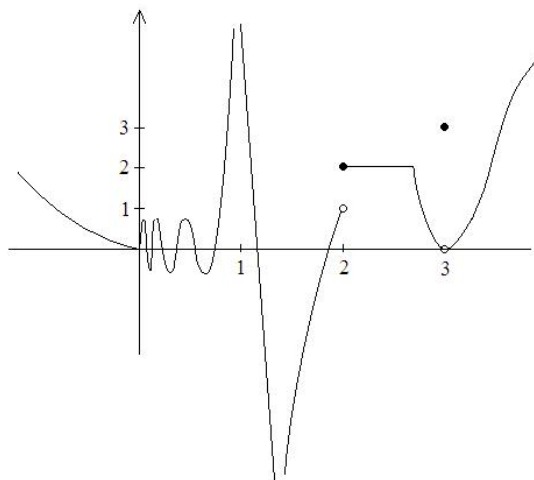


- i. $x \rightarrow a$, x 愈來愈靠近 a , 但從來不是 a .
- ii. • 極限存在 $\Rightarrow$ 極限唯一 $\Rightarrow$ 左極限 = 右極限.
• 左、右極限分別存在且相等 $\Rightarrow$ 極限存在.
- iii. • L 不一定是 $f(a)$.
- iv. If

$$f : \begin{cases} 1. \text{Polynomials} \\ 2. \text{Rational functions, } a \in \text{Domain}(f) \\ 3. \text{The graph of } f \text{ has no break,} \end{cases}$$

$$\Rightarrow L = f(a).$$

Example 1 :



(1) 那些 a 使得上述左、右極限，分別存在。

Solution : (i) $R - \{1\}$ (也即只有 $a = 1$ 時，左極限不存在).

(ii) $R - \{0, 1\}$ (也即在 $a = 0$ or 1 時，右極限不存在).

(2) 那些 a 使得上述極限存在。

Solution : $R - \{0, 1, 2\}$

(3) 那些 a 使得 $L = f(a)$.

Solution : $R - \{0, 1, 2, 3\}$

註:

$$\lim_{x_n \rightarrow 0} \sin \frac{1}{x_n} = \lim_{x_n \rightarrow 0} \sin \theta = \sin \theta.$$

$$\text{Here } x_n = \frac{1}{2n\pi + \theta}, \quad \theta \in R.$$

此表示當 x 由右邊趨近 0 時，若趨近路徑不相同時，極限不同。

Example 1 : $\lim_{x \rightarrow 1} \frac{x-1}{x^2-1} = ?$

Solution : $\lim_{x \rightarrow 1} \frac{x-1}{x^2-1} = \lim_{x \rightarrow 1} \frac{1}{x+1} = \frac{1}{2}$

Example 2 : $\lim_{x \rightarrow 0} (x^2 + 1) = ?$

Solution : $\lim_{x \rightarrow 0} (x^2 + 1) = 1$

Example 3 : $\lim_{x \rightarrow 1.5} \frac{2x^2 - 3x}{|2x - 3|} = ?$

Solution :

$$\lim_{x \rightarrow 1.5} \frac{2x^2 - 3x}{|2x - 3|} = \lim_{x \rightarrow 1.5} \frac{x(2x - 3)}{|2x - 3|} = 1.5$$

$$\lim_{x \rightarrow 1.5} \frac{2x^2 - 3x}{|2x - 3|} = \lim_{x \rightarrow 1.5} \frac{x(2x - 3)}{-(2x - 3)} = -1.5$$

$$\Rightarrow \lim_{x \rightarrow 1.5} \frac{2x^2 - 3x}{|2x - 3|} \text{ 不存在}$$

Example 4 : $\lim_{t \rightarrow 0} \frac{\sqrt{t^2 + 9} - 3}{t^2} = ?$

Solution :

$$\begin{aligned} & \lim_{t \rightarrow 0} \frac{(\sqrt{t^2 + 9} - 3)(\sqrt{t^2 + 9} + 3)}{t^2 (\sqrt{t^2 + 9} + 3)} \\ &= \lim_{t \rightarrow 0} \frac{1}{\sqrt{t^2 + 9} + 3} \\ &= \frac{1}{6} \end{aligned}$$

Example 5 : $\lim_{x \rightarrow 1} \left[\frac{1}{x-1} - \frac{2}{x^2-1} \right] = ?$

Solution : $\lim_{x \rightarrow 1} \left(\frac{x-1}{x^2-1} \right) = \lim_{x \rightarrow 1} \frac{1}{x+1} = \frac{1}{2}$

Example 6 : $\lim_{t \rightarrow 0} \left(\frac{1}{t\sqrt{1+t}} - \frac{1}{t} \right) = ?$

Solution :

$$\begin{aligned} & \lim_{t \rightarrow 0} \frac{(1 - \sqrt{1+t})}{t\sqrt{1+t}} \\ &= \lim_{t \rightarrow 0} \frac{(1 - \sqrt{1+t})(1 + \sqrt{1+t})}{t\sqrt{1+t}(1 + \sqrt{1+t})} \\ &= \lim_{t \rightarrow 0} \frac{1}{\sqrt{1+t}(1 + \sqrt{1+t})} \\ &= \frac{-1}{2} \end{aligned}$$

Example 7 : $\lim_{x \rightarrow 0} \frac{(3+x)^2 - 9}{x} = ?$

Solution : $\lim_{x \rightarrow 0} \frac{6x + x^2}{x} = 6$

Example 8 : $\lim_{x \rightarrow 2^+} [x] = ? \quad \lim_{x \rightarrow 2^-} [x] = ?$

Solution : $\lim_{x \rightarrow 2^+} [x] = 2$; $\lim_{x \rightarrow 2^-} [x] = 1$

Example 9 : $\lim_{x \rightarrow -2} \frac{3x^2 + ax + a + 3}{x^2 + x - 2}$ exists. Find a and its limit.

Solution :

$$3(-2)^2 + a(-2) + a + 3 = 0$$

$$\Rightarrow a = 15$$

$$\lim_{x \rightarrow -2} \frac{3x^2 + 15x + 18}{x^2 + x - 2}$$

$$= \lim_{x \rightarrow -2} \frac{3(x+2)(x+3)}{(x+2)(x-1)}$$

$$= -1$$

Example 10 : $\lim_{x \rightarrow 4} f(x) = \text{exists.}$ $f(x) = \begin{cases} \sqrt{x-4} & x > 4. \\ 8-ax & x < 4. \end{cases}$ Find a .

Solution : 右極限 $= 0 =$ 左極限 $= 8 - 4a$

$$\Rightarrow a = 2$$

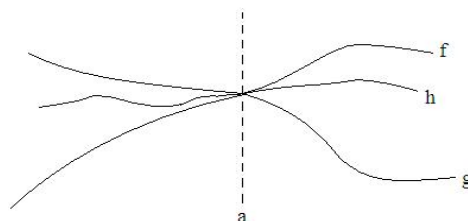
Theorem1. $\lim_{x \rightarrow a} f(x)$ exist $\Leftrightarrow \lim_{x \rightarrow a^-} f(x)$ and $\lim_{x \rightarrow a^+} f(x)$ exist and equal.

Theorem2. $\lim_{x \rightarrow a} f(x)$, $\lim_{x \rightarrow a} g(x)$ exist.

$$\Rightarrow \lim_{x \rightarrow a} f(x) * g(x) = \lim_{x \rightarrow a} f(x) * \lim_{x \rightarrow a} g(x)$$

Here $*$ $= +, -, \times$.

Theorem3. Sandwich Theorem.



If $f(x) \leq h(x) \leq g(x)$, except possibly at a , for x near a , and

$$\lim_{x \rightarrow a} f(x) = \lim_{x \rightarrow a} g(x) = L, \text{ then } \lim_{x \rightarrow a} h(x) = L.$$

Example 11 : $\lim_{x \rightarrow 0} x \sin \frac{1}{x} = ?$

Solution :

$$\begin{aligned} -|x| &\leq x \sin \frac{1}{x} \leq |x| \\ \Rightarrow \lim_{x \rightarrow 0} x \sin \frac{1}{x} &= 0 \end{aligned}$$

Example 12 : $\lim_{x \rightarrow 2} \frac{\sqrt{6-x}-2}{\sqrt{3-x}-1} = ?$

Solution :

$$\begin{aligned} &\lim_{x \rightarrow 2} \frac{\sqrt{6-x}-2}{\sqrt{3-x}-1} \\ &= \lim_{x \rightarrow 2} \frac{(2-x)(\sqrt{3-x}+1)}{(2-x)(\sqrt{6-x}+2)} \\ &= \frac{4}{2} \\ &= 2 \end{aligned}$$